BRAINSTORM GROUP (BSG) MAT111 TEST 2 (2021/22)

ANSWER ALL QUESTION

1. Evaluate $-\frac{4}{i^9}$ A. -4 B. 4 C. $-4i$ D. $4i$
2. Evaluate $-\frac{1}{i^7}$ A. $-i$ B. i C. 1 D. -1
3. Evaluate $\frac{2}{(1+i)^4}$ A. -2 B. 2 C. $-\frac{1}{2}$ D. $\frac{1}{2}$

Use $Z_1 = 1 + 2i$, $Z_2 = 4 - 3i$, $Z_3 = -2 + 3i$ to answer question 4-8.

4. $Z_1 + Z_2 - Z_3$
   A. $7 - 4i$ B. $-5 + 8i$ C. $3 + 2i$ D. $1 - 2i$
5. $Z_1 Z_3$
   A. $10 - 5i$ B. $-1 + 8i$ C. $-8 - i$ D. $1 - 2i$
6. $Z_2 - Z_1 + Z_3$
   A. $7 - 4i$ B. $-5 + 8i$ C. $3 + 2i$ D. $1 - 2i$
7. Evaluate $\frac{1-j}{1+j}$ A. $-j$ B. j C. 1 D. -1
8. $|Z_1 + Z_2| - |Z_3| \times \sqrt{2}$
   A. $\sqrt{10}$ B. 10 C. $2\sqrt{5}$ D. 0
9. Find $argZ\ if\ Z = i - 1$.
   A. $\frac{1}{2}\pi$ B. $\frac{2}{3}\pi$ C. $\frac{4}{3}\pi$ D. $\frac{3}{4}\pi$
10. The product of complex numbers results in ___ number.
    A. real B. imaginary C. complex D. natural
11. Given $(1 + 2i)(-2 - 3i) = a + bi$. Find $a + b$.
    A. -3 B. 3 C. -4 D. 4
12. Given $2 - 3i = \sqrt{a + bi}$. Find a and b.
    A. -5 and -12 B. 5 and -12
    C. -5 and 12 D. 5 and 12
13. If $(x - 2yi) - (y - xi) = 2 + i$, find x.
    A. -3 B. -1 C. 3 D. 1
14. If $\frac{2+j}{1-j} = j(x + yj)$. Find y and x.
    A. $\frac{1}{2}, \frac{3}{2}$ B. $\frac{3}{2}, \frac{1}{2}$ C. $\frac{1}{2}, -\frac{3}{2}$ D. $-\frac{1}{2}, \frac{3}{2}$
15. Given $A = 1 + 2i$, the conjugate of A is
    A. $-1 + 2i$ B. $-1 - 2i$ C. $1 - 2i$ D. $1 + 2i$
16. Determine the modulus, argument of $2 + 4i$.
    A. $2\sqrt{5},\ 63.43°$ B. $2\sqrt{10},\ 63.43°$
    C. $2\sqrt{5},\ 26.57°$ D. $2\sqrt{10},\ 26.57°$
17. Simplify $i^9 + 2i^{11} + i^{13}$.
    A. -4 B. -1 C. 1 D. 0
18. Simplify $i^3(1 + i) + i^5(3 - i) + i^7(2 + i)$
    A. 0 B. 2 C. 3 D. 4
19. ___ numbers ranges from $-\infty$ to $+\infty$
    A. real B. integer C. complex D. natural
20. Find the solution of the equation $x^2 + 2x = -5$
    A. $-1 \pm 2i$ B. $1 \pm 2i$ C. $-2 \pm i$ D. $2 \pm i$
21. Simplify $(i - 1)^4$. A. $-4i$ B. -4 C. $4i$ D. 4
22. The fourth term in the expansion of $(2 + \frac{3}{x})^8$ is
    A. $48348x^{-3}$ B. $48384x^{-3}$
    C. $48834x^{-3}$ D. $48483x^{-3}$
23. Obtain the 6th term in the expansion of $(2 + \frac{1}{2}x)^8$.
    A. $14x^4$ B. $14x^5$
    C. $70x^4$ D. $70x^5$
24. Obtain the coefficient of x^4 in $(x - y)^9$.
    A. -126 B. 126 C. $126y^5$ D. $-126y^5$
25. Find the 2nd term in the expansion of $(2x - 1)^2$.
    A. $-4x$ B. $4x$ C. $4x^2$ D. $-4x^2$
26. Calculate the coefficient of x^3y^2 in $(2x - 3y)^5$.
    A. 10 B. -108 C. 720 D. -720
27. What is the coefficient of x^3 in $(2 + 3x)^4$.
    A. 72 B. 108 C. 180 D. 216
28. Calculate a if the coefficient of x^3 in $(a + 2x)^5$ is 320.
    A. -2 B. 3 C. 4 D. -4
29. Obtain the first three terms of the expansion $(1 + x)^{1/m}$.
    A. $1 - \frac{1}{m}x - \frac{1}{2}(1 - m)\frac{x^2}{m^2}$
    B. $1 - \frac{1}{m}x - \frac{1}{2}(1 + m)\frac{x^2}{m^2}$
    C. $1 + \frac{1}{m}x - \frac{1}{2}(1 + m)\frac{x^2}{m^2}$
    D. $1 + \frac{1}{m}x - \frac{1}{2}(1 - m)\frac{x^2}{m^2}$
30. What is the coefficient of x^3 in $(x - \frac{1}{2})^8$.
    A. 7 B. -7 C. $\frac{7}{4}$ D. $-\frac{7}{4}$
31. Obtain the first three terms of the expansion $(1 + x)^{-m}$.
    A. $1 - mx - \frac{1}{2}m(m + 1)x^2$
    B. $1 + mx + \frac{1}{2}m(m - 1)x^2$
    C. $1 - mx + \frac{1}{2}m(m + 1)x^2$
    D. $1 + mx + \frac{1}{2}m(m - 1)x^2$
32. Find the term independent of x in the expansion of $(x^2 - \frac{2}{x})^6$.
    A. -16 B. 16 C. -240 D. 240
33. Find the term independent of x in the expansion of $(x - \frac{3}{x})^8$.
    A. 5760 B. 5670 C. -5670 D. -5760
34. If $(1 - 2x)^6 = 1 + Px + Qx^2$ ... find the values of the constants P and Q.

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A. $60\ and\ -12$ B. $-60\ and\ 12$
C. $12\ and\ -60$ D. $-12\ and\ 60$

35. Express $\frac{(n+2)!}{n!}$
A. $(n+1)(n+2)$ B. $\frac{1}{2}(n+1)(n+2)$
C. $(n-1)(n-2)$ D. $n(n-1)(n-2)$

36. If $\frac{(2x+1)!}{(2x)!} = 9$, find the value of x.
A. 4 B. 4.5 C. 5 D. 6

37. In a class of 300 students, 200 register Mathematics and 150 register Computer Science. Find the number of students that register at least one of the courses if only 40 students register none.
A. 90 B. 150 C. 260 D. 300

38. Two sets, A and B, are said to be disjoint if
A. $A \cap B = 0$ B. $A \cup B = 0$
C. $A \cup B = \emptyset$ D. $A \cap B = \emptyset$

39. P and Q are two sets such that is $n(P \cup Q)$ is 50, n(P) is 40 and n(Q) is 32. What is n$(P \cap Q)$?
A. 10 B. 12 C. 18 D. 22

40. Given that $X = \{x: 3 < x < 6\}\ and\ Y = \{x: 4 \leq x \leq 6$. Find $X \cap Y$.
A. $\{4, 5\}$ B. $\{4, 5, 6\}$
C. $\{3, 4, 5\}$ D. $\{3, 4, 5, 6\}$

41. Given U $= \{Integers\ between\ 0\ and\ 10\}, P = \{x: x < 5\}\ and\ Q = \{6 \leq x < 8\}. Find\ (P \cup Q)'$
A. $\{1, 2, 3, 4\}$ B. $\{1, 2, 3, 4, 7, 8\}$
C. $\{5, 8, 9\}$ D. $\{0. 5, 9, 10\}$

42. Given that U $= \{x : x$ is an integer, $2 \leq x \leq 10\}, A = \{x : x\ is\ a\ prime\ number\}, B = \{x:\ is\ a\ multiple\ of\ 3\}. Find\ (A \cup B)'$.
A. $\{4, 8\}$ B. $\{2, 3, 5\}$
C. $\{4, 8, 10\}$ D. $\{2, 3, 5, 6, 7, 9\}$

43. Given the Universal set $\{U = x : x \in \mathbb{N}, x < 15\}$ with its subsets as A = {odd numbers}, B = {perfect squares} and C = {numbers divisible by 3}. List elements of $(A' \cap B') \cap C$.
A. $\{0, 6, 10\}$ B. $\{2, 8, 10, 14\}$
C. $\{6, 12\}$ D. $\{2, 8, 10, 14\}$

44. Evaluate $\sum_{r=1}^{5} r(2r + 3)$.
A. 45 B. 90 C. 145 D. 155

45. Evaluate $\sum_{r=1}^{4} r(r^2 + 1)$.
A. 100 B. 110 C. 120 D. 130

46. The 4th term of an A.P. is 6. If the sum of the 8th and 9th terms is -72, the common difference is
A. $-\frac{28}{3}$ B. $-\frac{28}{6}$ C. $-\frac{28}{9}$ D. 34

47. The first term of an A.P. is 6 and the fifth term is 18. Find the number of terms in the series having a sum of 162.
A. -12 B. 12 C. 6 D. 9

48. The second and fifth terms of a G.P. are $\frac{2}{3}$ and $-\frac{1}{12}$ respectively. The first term is
A. $\frac{3}{4}$ B. $\frac{4}{3}$ C. $-\frac{4}{3}$ D. $-\frac{3}{4}$

49. Obtain the sum of positive even integers less than 100.
A. 4900 B. 5050
C. 5500 D. 2450

50. Resolve into partial fraction $\frac{2x^2+3x+10}{x^2+3x+2}$.
A. $2 + \frac{9}{(x+1)} + \frac{12}{(x+2)}$
B. $2 - \frac{9}{(x+1)} + \frac{12}{(x+2)}$
C. $2 + \frac{9}{(x+1)} - \frac{12}{(x+2)}$
D. $2 - \frac{9}{(x+1)} - \frac{12}{(x+2)}$

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ANSWERS									
1	D	11	A	21	B	31	C	41	C
2	A	12	A	22	B	32	D	42	C
3	C	13	C	23	B	33	B	43	C
4	A	14	D	24	D	34	D	44	D
5	C	15	C	25	A	35	A	45	B
6	D	16	A	26	C	36	A	46	A
7	A	17	D	27	D	37	C	47	D
8	D	18	C	28	A	38	D	48	C
9	D	19	B	29	D	39	D	49	D
10	C	20	A	30	D	30	A	50	C

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